CHAPTER ONE

INTRODUCTION

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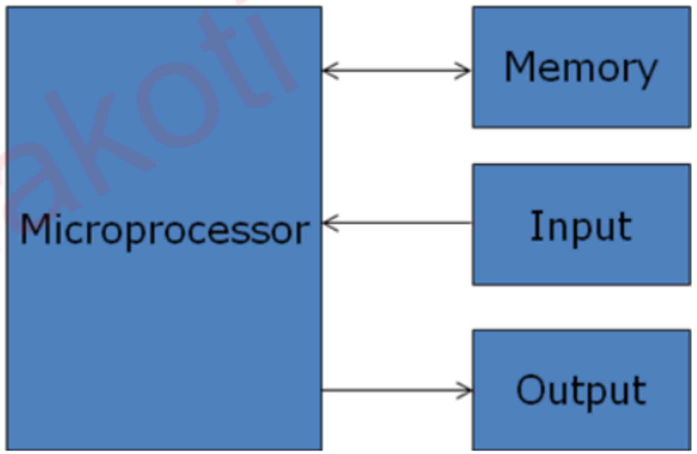
Microprocessor

- Multipurpose, programmable, clock driven, register based integrated circuit fabricated using LSI, VLSI etc. technology.
- Has limited set of onchip memory location called registers to hold information
- Clock synchronizes all the components present in microprocessor
- Can understand fixed set of basic commands and can generate signals to control external devices
- When interfaced with other devices, it can read instructions from memory, accepts data from input devices and process data according to the instructions and provide output
- Communicates with the peripherals (memory, I/Os) using system bus



- **CPU**: - Central processing unit which consists of ALU and control unit.
- **Microprocessor**: - Single chip containing all units of CPU.
- **Microcomputer**: - Computer having microprocessor as CPU.
- **Microcontroller**: single chip consisting of MPU, memory, I/O and interfacing circuits.
- **MPU**: - Microprocessing unit – complete processing unit with the necessary control signals.

History of Microprocessor and Computer

- Assignment

Basic Block Diagram of a Computer

- The CPU contains various registers to store data, the ALU to perform arithmetic and logical operations, instruction decoders, counters and control lines.

- The CPU reads instructions from memory and performs the tasks specified. It communicates with input/output (I/O) devices either to accept or to send data, the I/O devices is known as peripherals.

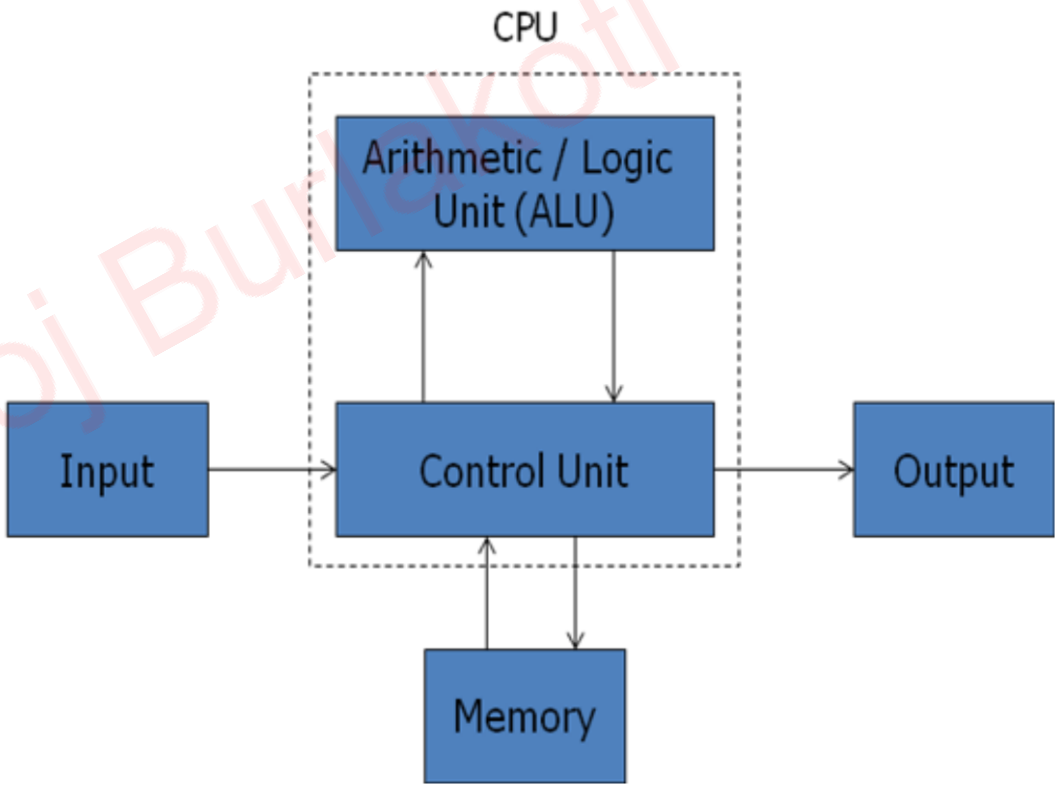


Fig: Traditional block diagram of a Computer

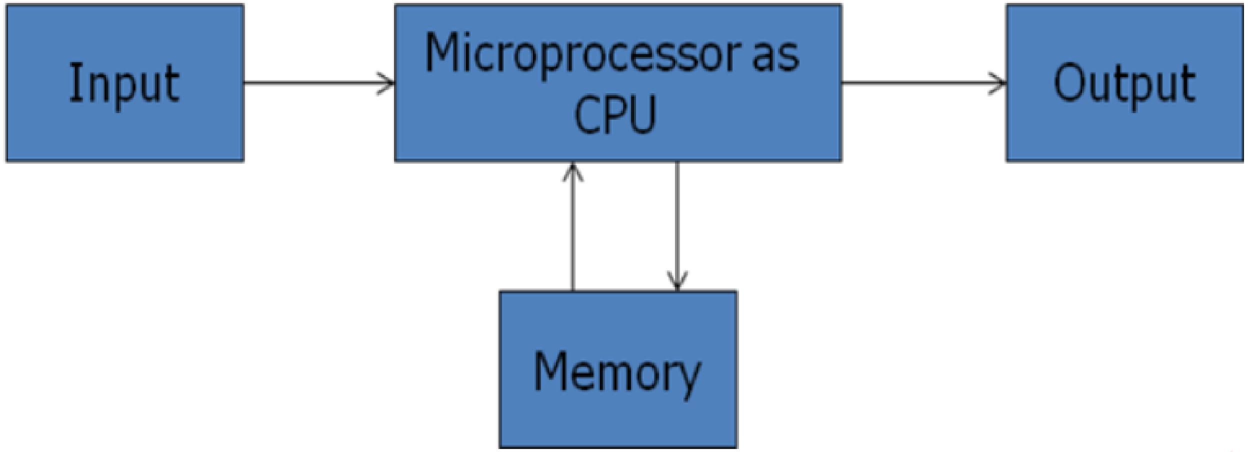


Fig: Block Diagram of a computer with the Microprocessor as CPU

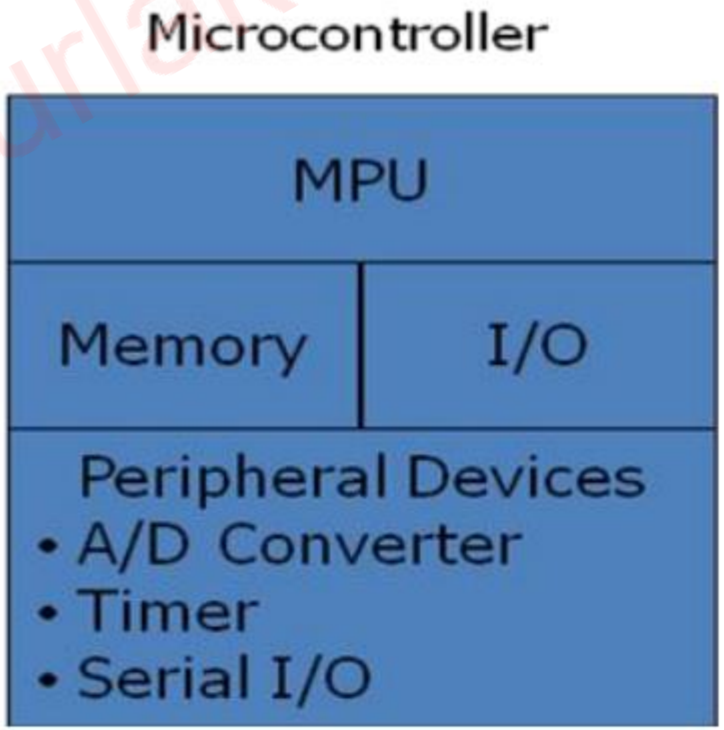


Fig: Block Diagram of a Microcontroller

Organization of a microprocessor based system

- Microprocessor based system includes there components microprocessor, input/output and memory (read only and read/write).

- These components are organized around a common communication path called a bus.

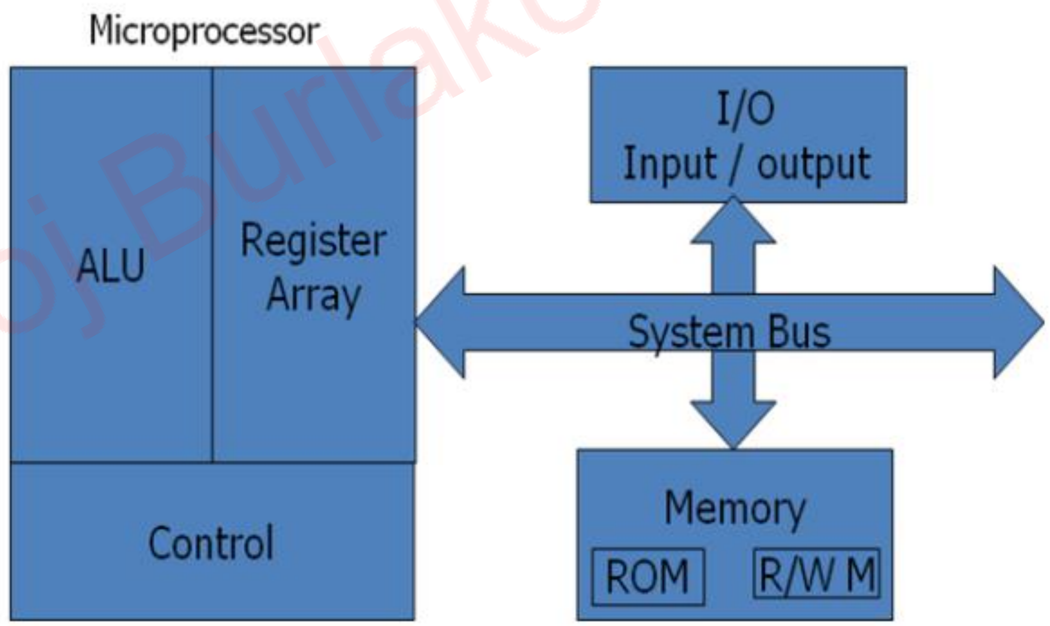


Fig: Microprocessor Based System with Bus Architecture

1. Microprocessor

A. Arithmetic/Logic unit
   - It performs arithmetic operations as addition and subtraction and logic operations as AND, OR & XOR.

B. Register array
   - The registers are primarily used to store data temporarily during the execution of a program and are accessible to the user through instruction.
   - The registers can be identified by letters such as B, C, D, E, H and L.

C. Control unit
   - It provides the necessary timing and control signals to all the operations in the microcomputer.
   - It controls the flow of data between the microprocessor and memory & peripherals.

2. Memory

- Memory stores binary information such as instructions and data, and provides that information to the up whenever necessary.
- To execute programs, the microprocessor reads instructions and data from memory and performs the computing operations in its ALU. Results are either transferred to the output section for display or stored in memory for later use. Memory has two sections.

A. Read only Memory (ROM):

- Used to store programs that do not need alterations and can only read.

B. Read/Write Memory (RAM):

- Also known as user memory which is used to store user programs and data.
- The information stored in this memory can be easily read and altered.

3. Input/Output

- It communicates with the outside world using two devices input and output which are also Known as peripherals.
- The input device such as keyboard, switches, and analog to digital converter transfer binary information from outside world to the microprocessor.
- The output devices transfer data from the microprocessor to the outside world. They include the devices such as LED, CRT, digital to analog converter, printer etc.

4. System Bus

- It is a communication path between the microprocessor and peripherals; it is nothing but a group of wires to carry bits.

Bus Organization

- Bus is a common channel through which bits from any sources can be transferred to the destination.
- A typical digital computer has many registers and paths must be provided to transfer instructions from one register to another.
- The number of wires will be excessive if separate lines are used between each register and all other registers in the system.
- A more efficient scheme for transferring information between registers in a multiple register configuration is a common bus system.
- A bus structure consists of a set of common lines, one for each bit of a register, through which binary information is transferred one at a time.
- Control signals determine which register is selected by the bus during each particular register transfer.

- A very easy way of constructing a common bus system is with multiplexers.
- The multiplexers select the source register whose binary information is then pleased on the bus.
- A system bus consists of about 50 to 100 of separate lines each assigned a particular meaning or function.
- Although there are many different bus designers, on any bus, the lines can be classified into three functional groups; data, address and control lines.
- In addition, there may be power distribution lines as well.

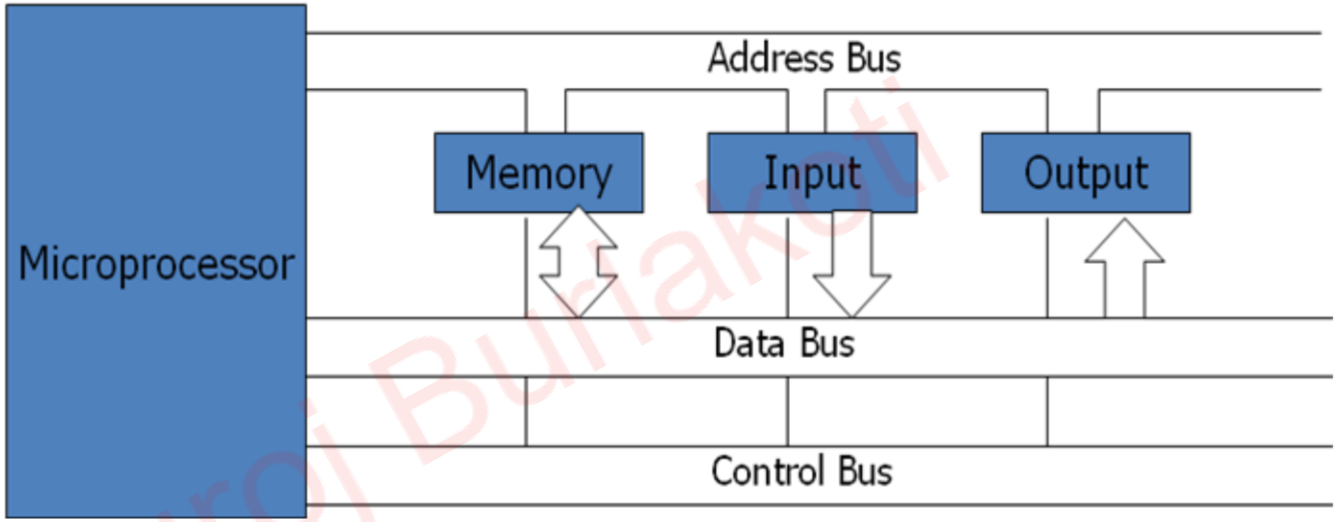


Fig: Bus Organization

Bus organization

- The **data lines** provide a path for moving data between system modules. These lines are collectively called data bus.
- The **address lines** are used to designate the source/destination of data on data bus.
- The **control lines** are used to control the access to and the use of the data and address lines. Because data and address lines are shared by all components, there must be a means of controlling their use. Control signals transmit both command and timing signals indicate the validity of data and address information. Command signals specify operations to be performed. Control lines include memory read/write, i/o read/write, bus request/grant, clock, reset, interrupt request/acknowledge etc.

Stored Program Concept and Von-Neumann Machine

- Rather than rewiring computer for each new task, in the stored program concept the program/instructions are stored in memory along side the data
- The program could be programmed or altered by changing the binary values
- This concept gives rise to sequential program execution becoming software programmable rather than hardware programmable
- This approach is known as 'stored- program concept' was first adopted by John Von Neumann and such architecture is named as von-Neumann architecture and shown in figure.

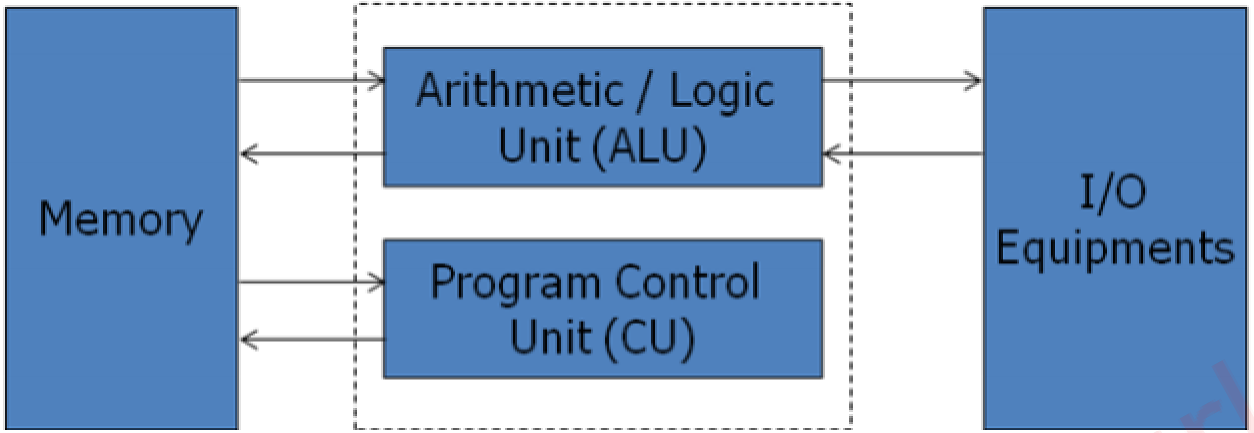


Fig: Von –Neumann Architecture

- The **main memory** is used to stare both data and instructions.
- The **arithmetic and logic unit** is capable of performing arithmetic and logical operation on binary data.
- The program **control unit** interprets the instruction in memory and causes them to be executed.
- The **I/O** unit gets operated from the control unit.
- The Von–Neumann architecture is the fundamental basis for the architecture of modern digital computers.
- It consisted of 1000 storage locations which can hold words of 40 binary digits and both instructions as well as data are stored in it.

- The storage location of control unit and ALU are called registers and there are various models of registers.

- MAR – memory address register – contains the address in memory of the word to be written into or read from MBR.
- MBR – memory buffer register – consists of a word to be stored in or received from memory.
- IR – instruction register – contains the 8-bit op-code instruction to be executed.
- IBR – instruction buffer register – used to temporarily hold the instruction from a word in memory.
- PC - program counter - contains the address of the next instruction to be fetched from memory.
- AC & MQ (Accumulator and Multiplier Quotient) - holds the operands and results of ALU after processing.

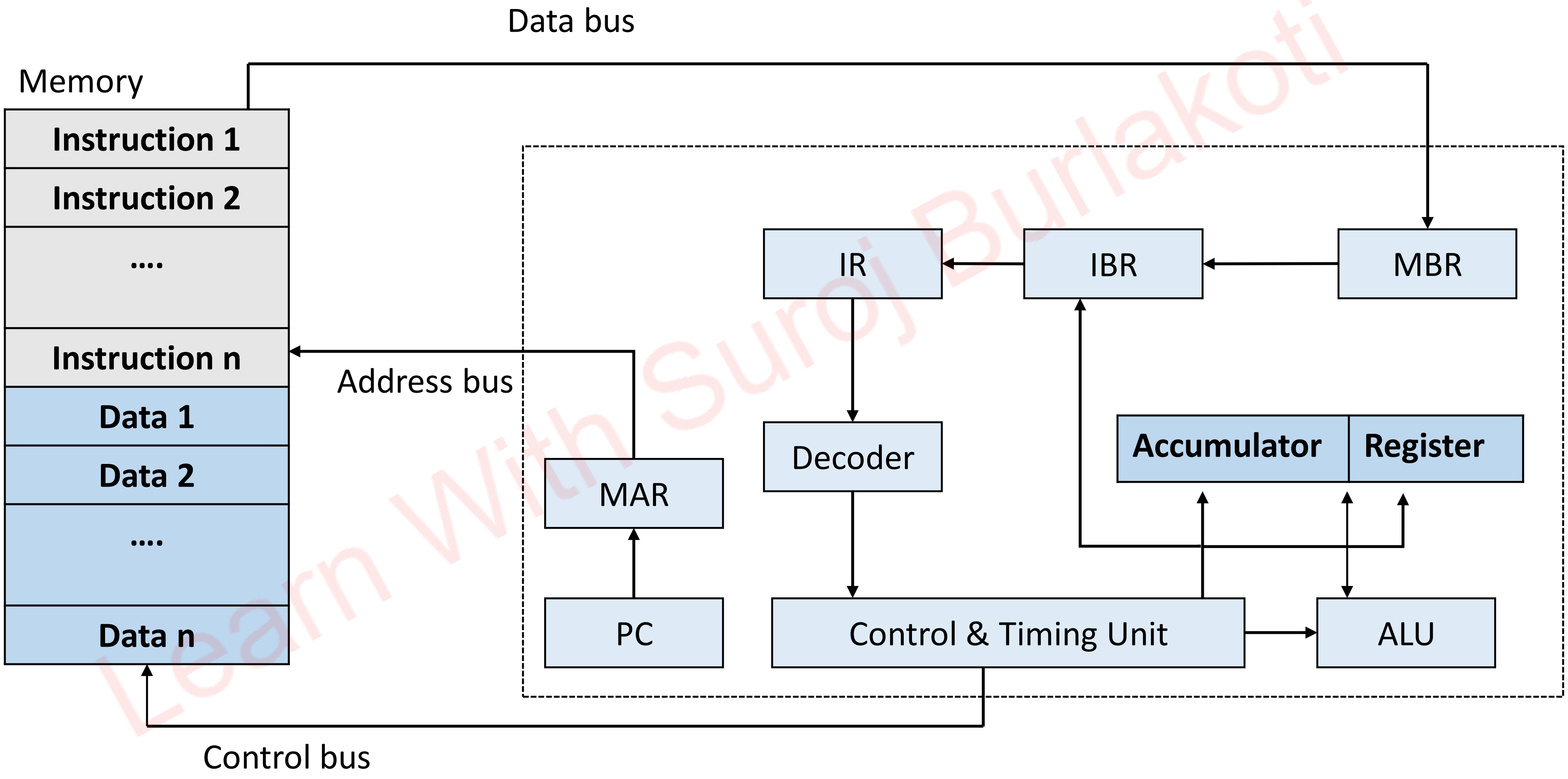


Fig: Structure of Von-Neuman’s Machine or IAS Computer

Stored Program Concept (operation)

- PC holds the address of recent executing instruction. MAR selects the particular memory location and the instruction is loaded to IR and PC is increased by One.
- The decoder decodes the instruction to control and timing unit
- The processing occurs at ALUand results are stored inappropriate registers

Harvard Architecture

- There is separate memory space for instructions (programs) and data with each memory space having its own data and address bus.
- So concurrent fetching of instruction and data from memory is possible, increasing the processing speed.
- It is used in microcontroller

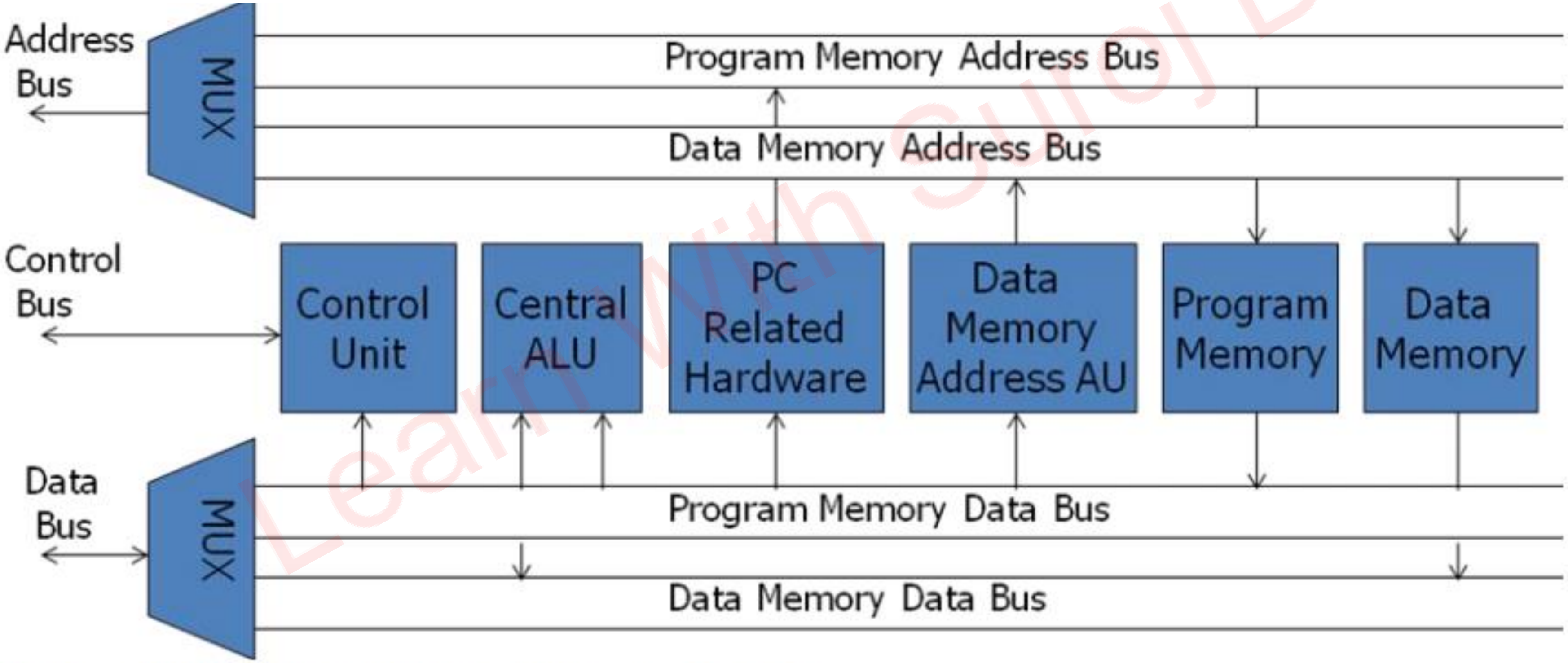


Fig: Harvard Architecture Based Microprocessor

Processing Cycle of a Stored Program Computer (Instruction Cycle)

- The processing cycle is the time required to complete the execution of one instructions
- Each cycle consist of two sub-cycles: Fetch Decode and Execute

1. Fetch decode
   - It fetches the instruction code from memory and decodes them
2. Execute
   - It executes the instruction which consist of calculating the address of operands, fetching them, performing operations on them and outputting the results to a specified location

Instruction Cycle

- While one instruction cycle is completed the cycle repeats itself beginning with a new instruction phase cycle
- Eg. MOV A,B

A. Fetch Decode Cycle

- Before microprocessor begins to execute the instruction it needs to fetch the code from memory where it is stored.

1. PC contains address of the next instruction to be executed. The content of PC is transfer to MAR.

T1: MAR ← PC

Fetch Decode Cycle

2. When control unit issues memory read signal, the content of memory locations specified by memory address register (MAR) is transfer into MBR.

T2: MBR ← [MAR]

3. Finally the content of MBR is transfer to IR and PC gets incremented by 1.

T3: IR ← MBR , PC← PC+1

Here IR contains the instruction code for execution which is then decoded after fetching

B. Execute Cycle

1. The address of B is transferred to MAR then to address bus.

   T1: MAR ← IR (address of B)

2. When control unit issues memory read signal, the content of B is transfer to MBR.

   T2: MBR ← B

3. The address of A from the address field IR is transferred to MAR.

   T3: MAR ← IR (Address of A)

4. When control unit issues memory write signal the contents of MBR is transferred to the memory location specified by MAR which is A in this case

   T4: A ← MBR

In the above case, the elementary operations perform on the information stored in one or more register within single unit of time is called micro-operation. These are smaller blocks of fetch decode and execute cycle.

Microinstructions

- Each instruction is characterized with many machine cycles and each cycle is characterized with many T-states.
- The lower instruction level patterns which are the numerous sequences for a single instruction are known as microinstructions.

Control Unit

1. Hardwired Control Unit

- The control unit essentially a state machine circuit design using conventional logic circuit design
- It's i/p logic signals are transformed into set of o/p logic signals which are control signals.
- The CU performs different operations in the basis of op-codes.
- We have to derive the Boolean expression for each control signal as a function of input.

It uses fixed instruction, fixed logic blocks of logic gate arrays, encoder, decoder, etc.

- Since modern processor needs a Boolean equation, it is very difficult to build a combinational circuit that satisfies all these operations.
- It has faster mode of operation.
- A hardwired control unit needs rewiring if design has to be modified.

Eg. Intel 8085, Motorolla 6802 etc

Control Unit

- 2. Micro-programmed Control Unit
- An alternative to hardwired CU.
- In micro-programmed control unit, the control information is stored in control memory.
- The control memory is programmed to initiate required sequence of operations.
- Use sequences of instructions to perform control operations performed by micro
- operations.
- Control address register contains the address of the next microinstruction to be read
- As it is read, it is transferred to control buffer register.
- Sequencing unit loads the control address register and issues a read command.
- It is cheaper and simple than hardwired CU.
- It is slower than hardwired CU.

Introduction to Register Transfer Language (RTL)

Thank You !